

MOHAMED CHERIF BRAHAM

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EDUCATION

Duke University, Durham, NC

Expected May 2029

B.S. in Electrical and Computer Engineering and Computer Science / Certificate in Robotics and Automation / GPA: 3.74

Coursework: Computer Architecture, Data Structures & Algorithms, Fundamentals of ECE, Eng. Design & Communication

UWC Adriatic, Duino, Trieste, Italy

Graduated May 2025

SKILLS

Programming: Python, C/C++, Java, JavaScript/TypeScript, Arduino

Hardware & Robotics: Arduino, Raspberry Pi, ESP32, Saeed Studio boards, Embedded Systems, PCB Design (KiCad), Sensor Integration, Forward/Inverse Kinematics, CAD (SolidWorks, OnShape), 3D Printing, CNC/Laser/Waterjet Cutting

AI/ML & Computer Vision: PyTorch, TensorFlow, YOLO/YOLOE, Pose Estimation, NLP (FastText), Edge AI / TinyML

Tools: Git, Linux, Next.js, ROS2, Supabase, AI-assisted dev (Claude), Docker, PostgreSQL, React, Hugging Face, FastAPI

Languages: Arabic (Native), French (Fluent), Italian (Proficient), English (Fluent)

RESEARCH & WORK EXPERIENCE

Duke Biomechanics and Wearable Robotics Lab, Duke University

2026 – Present

Research Assistant (Faculty Mentor: Dr. Krithika), Durham, NC

- Developing an OpenExo-based ankle exoskeleton for gait rehabilitation. Building skeleton and integrating sensing and control specification.
- Determined mechanical and technical requirements for the exoskeleton, including motor torque specifications, then reconciled three overlapping vendor BOMs against those requirements into a single purchasing document.
- Selected and validated IMU-based gait-event sensing against published biomechanics literature.

Innovation Co-Lab, Duke University (OIT)

Feb 2026 – Present

Makerspace Technician, Durham, NC

- Operating and maintaining 30+ 3D printers, waterjets, and laser cutters across Duke's three Innovation Co-Lab fabrication facilities.
- Providing daily hands-on technical guidance to 20+ students, faculty, and staff on effective and safe use of fabrication tools.
- Resolving recurring 3D-printer failures (jammed filament, clogged nozzles, filament changes); teaching students to safely operate the waterjet and laser cutters.

Independent Research – Robotic Manipulation & Kinematics (SO-100 Robotic Arm)

Aug – Sep 2026

Self-directed

- Built and deployed a fully autonomous, zero-shot pick-and-place system on the physical SO-100 arm: a Claude agent (via MCP) interpreted the camera feed and estimated target position; I built the surrounding system: MCP integration, camera calibration, and the IK control loop linking Claude's output to arm motion. Achieved repeated success (blue cube into bucket), no task-specific training; benchmarked Haiku/Sonnet/Opus: Opus fastest, most reliable.
- Separately developing a custom on-arm vision pipeline: open-vocabulary detection (YOLOE-seg) for target localization plus ArUco-tag visual serving for gripper alignment as an alternative to agent-based control. Work in progress; not yet reliable.
- Derived rotation matrices and homogeneous transforms from first principles; implemented a closed-form planar IK solver exploiting the arm's parallel joint geometry. Verified against forward kinematics in simulation over 300 round-trip trials, yielding near-zero residual error; simulation-only, not tested on hardware.

ATAST Organization, Tunisia

Nov 2020 – Aug 2023

AI and Robotics Instructor

- Instructed 15+ students in Python, Arduino, and computer vision (Yolo); supervised engineering-fair projects
- Supervised 2 middle school teams in Vex Robotics competition; Guided the build process

SELECTED TECHNICAL PROJECTS

do_robotics – Phone-as-Robot-Controller Platform

- Achieved real-time visual tracking and following of a physical target (a bottle) on a minimal platform: ESP32, battery, and 2 motors only, no onboard compute.
- Ran all vision/ML processing on-device via edge ML (TFLite) on a smartphone (Redmi Note 11), sending commands to the ESP32 over Bluetooth/USB-serial via a custom 5-byte protocol.
- Designed and 3D-printed a custom chassis to house the phone and onboard electronics.
- Built a block-based visual scripting interface within the companion app for behavior programming.

Blood Loss Monitoring Device for PAS Surgery

- Designed and silicone-molded a fluid-collection cup, pumping mechanism, and wiring harness for a real-time blood-loss measurement system.
- Validated to within 5 mL absolute error (<0.7% relative error) across a tested range of 250–1000 mL. System used clinically during cesarean hysterectomy (PAS surgery).

ImpaQ – Smart Impact-Sensing Mouthguard

- Built a wearable impact sensor (ESP32, accelerometer) interfaced with a Flutter mobile app for real-time TBI-risk monitoring; completed working prototype. (C++, Flutter)
- Diagnosed a Bluetooth communication failure on the board: insufficient power delivery from the onboard voltage regulator. Used a multimeter to isolate the fault, then replaced the regulator with one properly sized for the Bluetooth module, restoring reliable wireless communication.

Earlier Independent CV/Robotics Projects – WildGuard & Oil-Detecting UAV

- WildGuard (Wildfire Detection, Oct 2021 – Apr 2022):** Built a real-time CV/ML pipeline (YOLO on Raspberry Pi) fused with environmental sensor data and an end-to-end Python/Pandas time-series pipeline for wildfire detection and prediction; achieved ~83% detection accuracy across varying lighting conditions and distances.
- Oil-Detecting UAV (Autonomous Spill Response, Jun 2023):** Built a computer-vision pipeline (YOLO, Raspberry Pi) for real-time oil-spill detection from the vehicle's onboard camera, paired with closed-loop steering control to autonomously navigate toward spill locations.

AWARDS & LEADERSHIP

- Duke University International Scholar (2025–2029)
- STEM competition awards: Nat'l Rep., Expo Science Asia Dubai (2022); Gold Medal & Top-10, I-FEST² (2023); Bronze Medal, Genius Olympiad NY (2023); Gold Medal, I Giovani e le Scienze (2024); APA Award (2024)